LINKING FACEBOOK ENGAGEMENT TO SALES: A DECISION SUPPORT SYSTEM FOR TARGETED MARKETING AT PC MERCHANDISE

A Capstone Project by

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CHAPTER III

DESIGN AND METHODOLOGY

3.1 Requirement Analysis and Documentation

This section presents the comprehensive requirements for the proposed system, Linking Facebook Engagement to Sales: A Decision Support System for Targeted Marketing at PC Merchandise. The requirements define both the functional capabilities the system must deliver and the quality attributes necessary for the system's effective, reliable, and secure operation.

Requirements were gathered through a structured combination of interviews, direct observation, document analysis, and dataset profiling. These methods collectively ensured that the system design is grounded in the actual operational needs and data realities of PC Merchandise, and that the system adequately addresses the client's challenges in linking Facebook engagement activity to measurable sales outcomes. The analytical methods specified in this chapter were further determined through empirical profiling of the actual datasets provided by the client, ensuring that all chosen techniques are statistically appropriate for the nature and volume of the available data.

3.1.1 Requirements Gathering Methods

The researchers employed four complementary data-gathering methods to ensure that the system requirements accurately reflect the operational context of PC Merchandise. Table 1 summarizes each method, the purpose, and the corresponding participants or sources involved.

Table 1. Requirements Gathering Methods

Method	Purpose	Participants / Sources
Interview	To understand the roles, decision-making processes, and current challenges faced by key stakeholders in marketing and sales operations, and to identify the client's expectations for system outputs and analytics.	Business Owner, Marketing Manager, Sales Director of PC Merchandise
Observation	To examine the existing workflow for managing Facebook engagement data, campaign performance monitoring, and sales recording at the client site, including how data is currently exported and reviewed.	PC Merchandise daily operations and marketing reporting activities
Document Analysis	To review and assess available data sources, including Facebook Insights CSV exports, Facebook Ads Manager	Facebook Insights CSV, Facebook Ads Manager CSV, internal sales and transaction records

	CSV exports, and internal sales records, to identify usable fields and assess data structure.	
Dataset Profiling	To evaluate the structure, completeness, statistical distribution, and analytical suitability of the datasets provided by the client, including normality testing to determine the appropriate correlation and regression methods for the system.	Facebook organic post engagement datasets and Facebook paid campaign datasets provided by PC Merchandise

Among these methods, dataset profiling was particularly consequential to the analytical design of the system. Shapiro-Wilk normality tests conducted on all key engagement variables confirmed significant non-normality across both datasets, with p-values below 0.001 and skewness values ranging from 2.4 to 5.7. These results directly informed the selection of Spearman Rank Correlation over Pearson Correlation, the adoption of Simple Linear Regression with a single empirically selected predictor, and the use of month-over-month trend analysis in place of ARIMA-based forecasting — all of which are discussed in detail in the functional requirements that follow.

3.1.2 Functional Requirements

Functional requirements specify the behaviors and capabilities that the system must provide to fulfill the system's intended purpose. Table 2 presents each requirement with its corresponding identifier and description.

Table 2. Functional Requirements

ID	Name	Description
FR-01	User Access Management	The system shall provide a complete user access management workflow consisting of three functions. First, authentication — users shall log in using a registered username and password validated server-side, with access denied and an appropriate error message displayed for unrecognized or incorrect credentials. Second, account management — the Business Owner is the only authorized user who may create, edit, and deactivate user accounts, with account creation requiring a username, initial password, and role assignment. Third, password reset — users shall be able to reset their password through a secure process accessible from the login screen, requiring identity verification before a new password can be set.
FR-02	Role-Based Access Control	The system shall restrict access to features and data based on the authenticated user's assigned role. The Marketing Manager is the sole data operator, responsible for uploading CSV files, managing content categorization, monitoring campaign performance, running simulations, and accessing the marketing dashboard. The Sales Director accesses engagement-to-purchase correlation results, what-if simulation, campaign performance rankings, and report generation. The Business Owner accesses the

		executive dashboard, trend analysis, simulation, executive reports, and user account management.
FR-03	Data Upload Interface	The system shall allow the Marketing Manager to upload two types of manually exported CSV files — Facebook Insights data and Facebook Ads Manager data. The system shall automatically identify the file type based on the uploaded file's column structure. The system does not connect to Facebook directly in any form.
FR-04	Data Validation	The system shall validate all uploaded CSV files for correct structure, required field presence, appropriate data types, and value consistency prior to storage and processing. Validation errors shall be reported to the user with specific descriptions of the detected issues.
FR-05	Data Storage	The system shall store all validated and processed data in a structured MySQL relational database organized to support efficient querying, retrieval, and analytical computation across all system modules.
FR-06	Data Preprocessing	The system shall preprocess uploaded datasets prior to storage by excluding irrelevant and entirely empty columns identified during dataset profiling, handling missing values based on each field's analytical purpose, On detecting duplicate records, the system will overwrite the existing record with the newly uploaded values to ensure engagement metrics always reflect the most recent export using a combination of ad name and reporting period as well as unique identifier for organic posts. The number of records inserted as new and the number overwritten will both be reported to the Marketing Manager through the upload validation results. Standardizing date formats, and computing engagement rate for each organic post record as the ratio of total reactions, comments, and shares to reach expressed as a percentage.

FR-07	Facebook Organic Post Data Processing	The system shall process post-level engagement data from Facebook Insights CSV files uploaded by the Marketing Manager, parsing and storing the following fields: Post ID, publish time, post type, title, description, permalink, reach, impressions, reactions, comments, shares, and views.
FR-08	Facebook Campaign and Ads Data Processing	The system shall process ad-level performance data from Facebook Ads Manager CSV files uploaded by the Marketing Manager, parsing and storing the following fields: ad name, ad set name, reporting period, reach, impressions, link clicks, amount spent, total messaging contacts, purchases, results, and cost per result. Records with missing values in total messaging contacts or purchases shall be retained in storage but excluded from correlation and regression computations.
FR-09	Data Aggregation and Alignment	The system shall organize stored organic post data and ads data to support both independent and comparative analysis. Both datasets shall be analyzed independently at the record level. For trend analysis, both datasets shall be aggregated by reporting month to enable month-over-month comparison across the same periods.
FR-10	Content Categorization Module	The system shall provide a module for categorizing both organic posts and paid ads by content type to enable performance comparison across categories. Predefined content categories and initial keyword sets are established by the researchers prior to deployment. The Marketing Manager is the only authorized user who may add, edit, or remove keywords within existing categories after deployment. Category names are fixed and cannot be modified by any user. This module consists of two sub-functions: FR-10.1 – Categorization Interface: the system shall present uncategorized posts and ads in a filterable,

		searchable interface where the Marketing Manager manually assigns content categories, with the permalink displayed for each organic post as reference to the original Facebook content. FR-10.2 – Keyword-Assisted Suggestions: the system shall analyze the Title field of organic posts and the Ad name field of paid ads against stored keyword sets to generate automated category suggestions that the Marketing Manager may accept, modify, or override before saving.
FR-11	KPI Computation	The system shall compute and display the following KPIs: Engagement rate per post, reach performance index per post, cost per messaging contact per ad, cost per purchase per ad, and monthly performance summaries aggregating total reach, impressions, spend, messaging contacts, and purchases per reporting month. Cost per result, exported directly by Facebook Ads Manager, shall be stored and displayed without recomputation. KPIs dependent on partially available fields shall only be computed for records where the required fields are present, and the number of records included shall be displayed alongside each result.
FR-12	Analytics Pipeline	The system shall implement a sequential three-stage analytical pipeline. Stage 1 – Correlation Analysis: the system shall apply Spearman Rank Correlation between engagement metrics and outcome variables — total messaging contacts and recorded purchases — presenting results through a coefficient table and heatmap visualization. Stage 1 runs each time the user requests analysis and reflects all records currently stored in the system.

		Stage 2 – Regression Modeling: using the strongest predictor of purchases identified in Stage 1, the system shall apply Simple Linear Regression to model the relationship between that predictor and recorded purchases, presenting the regression equation, R-squared value, and residual plot. The regression model shall retrain automatically each time new ad data is uploaded. Stage 3 – What-If Simulation: using the regression equation from Stage 2, the system shall allow authorized users to input a hypothetical predictor value and generate a projected purchase estimate alongside a historical baseline comparison. Stage 3 is only available after Stage 2 has produced a valid trained model.
FR-13	Trend Analysis	The system shall perform month-over-month trend analysis across the available campaign periods, aggregating and comparing total reach, total impressions, total spend, total messaging contacts, total purchases, cost per messaging contact, and cost per purchase per reporting month. Results shall be presented through comparative charts and a summary table displaying absolute values and period-over-period percentage changes.
FR-14	Dashboard and Visualization	The system shall display role-appropriate interactive dashboards presenting charts, graphs, and summaries of organic post performance, campaign performance, KPI results, correlation findings, regression outputs, trend analysis results, and simulation results according to the accessing user's assigned role.
FR-15	Report Generation	The system shall generate structured analytical reports summarizing campaign performance, engagement-to-purchase relationships, KPI results, and model outputs, available for download by

		authorized users according to each user's assigned role.
FR-16	Campaign Performance Analysis	The system shall provide a ranked performance table of individual ads sortable by reach, impressions, amount spent, total messaging contacts, cost per messaging contact, purchases, and cost per purchase. Records missing the selected metric shall be excluded from that specific ranking without being removed from the system. Ads in the top 25 percent of the selected metric shall be labeled high-performing and those in the bottom 25 percent shall be labeled underperforming. The system shall also aggregate performance by content category assigned through FR-12, displaying average reach, average messaging contacts, total purchases, and average cost per messaging contact per category. Category-level aggregation is only available for ads that have been assigned a content category.
FR-17	System Deployment	The system shall be deployed on a dedicated local host machine at PC Merchandise using Cloudflare Tunnel, providing secure internet-accessible connectivity through encrypted HTTPS for all authorized users regardless of physical location, without requiring external cloud hosting infrastructure.

The functional requirements in Table 2 reflect decisions grounded in the actual operational context of PC Merchandise and the statistical properties of the datasets provided by the client.

On role design (FR-02), the Marketing Manager will be designated as the sole data operator because uploading CSV exports and managing post categories require direct familiarity with Facebook campaign content. The Sales Director will

function as an analytical consumer whose role is to interpret engagement-to-purchase outcomes. The Business Owner will hold executive access to strategic outputs and will be the appropriate authority for user account management since that is an administrative responsibility that should not rest with operational staff.

On the upload interface (FR-03), the system will not connect to Facebook in any form. All data will enter the system exclusively through CSV files manually exported by the Marketing Manager from Facebook Insights and Facebook Ads Manager.

On missing values in ads data (FR-08), records with missing values in total messaging contacts and purchases will be retained in storage but excluded from analytical computations because their absence reflects campaign optimization objectives. Facebook only records messaging contacts for ads configured with a messaging objective and only records purchases for ads with conversion tracking enabled. Treating these absences as data quality errors and removing the records entirely would significantly reduce the usable analytical sample.

On content categorization (FR-10), the predefined categories and initial keyword sets will be established by the researchers prior to deployment based on recurring content themes identified during dataset profiling of the 81 organic post records and 230 ad records provided by PC Merchandise. Category names will be fixed and will not be modifiable after deployment to ensure that performance comparisons across time periods remain consistent — changing a category name

mid-study would make it impossible to meaningfully compare category-level metrics across months. The permalink will be displayed alongside each organic post in the categorization interface because 44.4 percent of organic post records in the current dataset have no description field, meaning keyword matching cannot be performed for nearly half of all posts, requiring the Marketing Manager to refer to the actual post on Facebook before assigning a category manually. Upon acquisition of the complete eight-month dataset, the system will monitor uncategorized records after each upload and flag recurring unmatched content patterns that may warrant the creation of a new category, surfacing these findings to the Marketing Manager for review. This flag-and-suggest mechanism is a planned enhancement and is not part of the current implementation scope.

On the analytics pipeline (FR-12), the three stages will not function as independent features — each stage will depend on the output of the preceding one. Correlation will inform predictor selection, predictor selection will inform regression modeling, and the regression model will power the simulation. The use of Spearman Rank Correlation in Stage 1 is statistically mandated by the properties of the actual data. Shapiro-Wilk normality tests conducted on all key engagement variables produced p-values of 0.0000 with skewness values ranging from 2.4 to 5.7, confirming severe non-normality across both datasets. Pearson Correlation requires normally distributed data and produces unreliable coefficients when this assumption is violated. Akoglu (2021) establishes that Spearman correlation is the appropriate alternative when normality assumptions are not met,

particularly in applied research involving behavioral or engagement data. The selection of Simple Linear Regression in Stage 2 is justified by two independent findings from the dataset profiling conducted by the researchers. First, the current sample of 42 ad records with purchases supports only one predictor under the minimum observations-per-predictor convention. Knofczynski and Mundfrom (2022) confirm that regression models with insufficient observations per predictor produce unstable coefficients, particularly in samples below 50 observations. Second, Variance Inflation Factor analysis on all candidate predictors produced values ranging from 26.46 to 158.63, far exceeding the acceptable threshold of 5.0, confirming extreme multicollinearity that makes multiple predictors unreliable within a single model. Spearman correlation on the 42 purchase records identified amount spent as the strongest predictor at $r = 0.72$, producing the equation: Purchases = 1.8168 + 0.000705 × Amount Spent with an R-squared value of 0.2658. Zhou et al. (2021) applied Simple Linear Regression to model the relationship between YouTube advertising spend and company sales, obtaining an R-squared of 0.4366, and concluded that social media advertising spend is a statistically significant predictor of sales outcomes. Similarly, Lin et al. (2021) applied Simple Linear Regression to Facebook advertising spend and sales data and confirmed a significant positive relationship between the two variables. Both studies establish precedent for using advertising spend as a single predictor in a Simple Linear Regression model for social media sales analysis, supporting the methodological approach to be adopted in this study. Upon acquisition of the

complete eight-month dataset, the researchers will evaluate expanding the model to Multiple Linear Regression following VIF-based predictor screening. The regression model will retrain automatically following each successful ads CSV upload that adds new records containing purchase data. The updated regression equation, R-squared value, sample size, and timestamp of the last training will be stored in the database and displayed on the dashboard so users can see when the model was last updated. Before retraining, the system will verify that the total number of ad records with recorded purchases meets the minimum sample threshold. If the uploaded file contains no new purchase records, retraining will be skipped and the existing model will be retained.

On trend analysis (FR-13), month-over-month aggregation will be applied rather than time-series forecasting methods because the current dataset covers only three non-consecutive months with October and November 2025 absent. Time-series methods such as ARIMA require consecutive equally spaced observations — a condition the current dataset fails on both count and continuity. The module will be designed to accommodate extension to appropriate forecasting methods upon acquisition of the complete eight-month consecutive dataset.

On campaign performance rankings (FR-16), ads will be ranked by individual metrics rather than a composite weighted score because assigning weights to different metrics would require domain expertise and objective justification beyond the scope of this study. The top and bottom 25 percent thresholds for high-performing and underperforming labels will be based on

quartile boundaries derived from the actual distribution of each metric within the dataset, making the labels data-driven rather than arbitrary.

3.1.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints that govern how the system performs its functions. Table 3 presents each requirement with its identifier and description.

Table 3. Non-Functional Requirements

ID	Name	Description
NFR-01	Performance	The system shall complete all analytical computations — including preprocessing, KPI computation, Spearman Rank Correlation, Simple Linear Regression, and trend aggregation — and return results to the user within 10 seconds for datasets of up to 500 records under normal operating conditions. This benchmark accounts for Django request handling, MySQL database operations, and Cloudflare Tunnel network latency in addition to computation time.
NFR-02	Usability	The system shall provide a clean, role-appropriate web-based interface with clearly labeled navigation, consistent layout across all screens, and analytical outputs presented in a format interpretable without statistical expertise.
NFR-03	Security	The system shall store all user passwords using bcrypt hashing. All data transmitted between the user's browser and the server shall be encrypted via HTTPS enforced automatically through Cloudflare

		Tunnel. No plaintext passwords shall be stored or transmitted at any point.
NFR-04	Reliability	The system shall ensure data integrity throughout the upload, validation, preprocessing, and storage pipeline. No data shall be lost or corrupted during file processing, and all upload events shall be recorded in the system's upload log regardless of whether the upload succeeded or failed.
NFR-05	Accuracy	The system shall produce statistically accurate outputs for all analytical computations including Shapiro-Wilk normality tests, Spearman Rank Correlation coefficients and p-values, Variance Inflation Factor values, Simple Linear Regression equation and R-squared value, KPI calculations, and month-over-month trend aggregations, based on validated input data stored in the database.
NFR-06	Accessibility	The system shall be fully functional via modern web browsers including Google Chrome and Microsoft Edge without requiring the installation of additional software on the user's device.
NFR-07	Scalability	The system shall support the progressive addition of organic post records and ad records over time without degradation of system performance or analytical accuracy. The analytical pipeline shall accommodate the full eight-month dataset upon acquisition without requiring changes to the system architecture.
NFR-08	Maintainability	The system shall be developed using a modular architecture with clearly separated components for data ingestion, analytics processing, and presentation to allow independent updates, debugging, and future integration of additional analytical methods as the dataset expands.

NFR-09	Deployment	The system shall be deployable on a local host machine at PC Merchandise using Cloudflare Tunnel to provide secure, internet-accessible operation without third-party cloud hosting, ensuring availability for all role-based users from any location via a standard web browser.

The non-functional requirements presented in Table 3 are grounded in the specific operational context, deployment setup, and technical characteristics of the proposed system.

The 10-second performance benchmark defined in NFR-01 is established on two grounds. First, the analytical methods to be employed by the system — Spearman Rank Correlation, Simple Linear Regression, and month-over-month aggregation — are computationally lightweight statistical operations that do not involve iterative optimization or large matrix computations. Applied to datasets of up to 500 records, these methods will impose minimal processing demand on the host machine even without specialized hardware. The 10-second threshold is therefore a conservative target that allocates the majority of the allowed response time to web application overhead — specifically Django request handling, MySQL database read and write operations, and the network latency introduced by routing all traffic through Cloudflare Tunnel — rather than to the analytical computations themselves. Second, the 10-second threshold aligns with the established maximum acceptable web response time for maintaining user attention, as documented by Nielsen (2024) of the Nielsen Norman Group, who identifies 10

seconds as the upper limit beyond which users lose focus and the interaction is perceived as broken.

The scalability requirement in NFR-07 addresses two distinct dimensions of growth. The first is data volume growth within the current analytical approach — as the Marketing Manager uploads new monthly CSV exports, the number of organic post and ad records in the database will grow continuously. The planned use of MySQL with indexed queries and Python's pandas library for in-memory computation will ensure that this growth does not degrade analytical performance within the projected record volumes. The second dimension is methodological growth — when the complete eight-month dataset is acquired following proposal approval, the researchers will evaluate expanding the regression model from Simple Linear Regression to Multiple Linear Regression and extending the trend analysis module to time-series forecasting methods. The modular design of the analytics pipeline — where each analytical stage will be implemented as an independent Python module called by Django views — will ensure that expanding or replacing any analytical method requires changes only to the relevant module and not to the database schema, upload pipeline, or presentation layer.

3.1.4 Data Requirements

The system depends on structured datasets provided by PC Merchandise to perform meaningful analysis and generate reliable outputs. Table 4 presents the data requirements.

Table 4. Data Requirements

Category	Description
Data Sources	Facebook Insights post-level CSV exports and Facebook Ads Manager ad-level CSV exports, both manually exported by the Marketing Manager and uploaded through the system. Purchase counts from the Ads Manager export serve as the primary sales outcome variable.
Required Fields	Organic Post Data: Post ID, publish time, post type, title, description, permalink, reach, impressions, reactions, comments, shares, views. Engagement rate is computed during preprocessing and stored as a derived field. Ads Data: Ad name, ad set name, reporting starts, reporting ends, reach, impressions, link clicks, amount spent, total messaging contacts, purchases, results, cost per result, attribution setting.
Current Dataset	81 organic post records from September 2025 and 230 ad-level records from September 2025, December 2025, and January 2026, of which 130 contain messaging contact data and 42 contain recorded purchases.
Planned Full Dataset	Eight consecutive months of organic post and campaign data to be provided by PC Merchandise upon proposal approval.
Time Coverage	Current: September 2025 for organic posts and three non-consecutive months for ads. Planned: January 2025 to August 2025 consecutively.

Data Quality Standards	Datasets must be exported directly from Facebook platforms in CSV format. Entirely empty columns shall be excluded from storage. Records with missing values in analytical fields shall be excluded from computations but retained in storage. Duplicates shall be identified using ad name and reporting period.
Data Access Authorization	All datasets were provided by PC Merchandise under explicit authorization for academic use and a Non-Disclosure Agreement.

The data sources for the proposed system will be limited to two CSV export types — Facebook Insights and Facebook Ads Manager — both of which will be manually exported by the Marketing Manager and uploaded through the system's upload interface. The system will not connect to Facebook through any API or automated integration. There will be no Point-of-Sale system integration and no separate sales file. The system will use Facebook Ads Manager purchase counts as the sales outcome variable. Facebook Ads Manager records purchases attributed to ads within a seven-day click or one-day view window as specified in the attribution setting field of the exported CSV, representing the most direct and verifiable link between advertising activity and purchase outcomes available in the provided data.

The required fields listed in Table 4 reflect only the columns that will be retained after preprocessing. The raw Facebook Insights CSV contains 35 columns, of which 12 will be retained as analytically relevant. The raw Ads Manager CSV contains 33 columns, of which 13 will be retained. All excluded

columns fall into one of three categories — entirely empty across all records, operational configuration fields with no analytical value, or metrics redundant with retained fields or applicable only to a subset of post or campaign types.

The distinction between the current and planned datasets is the most consequential data requirement for the integrity of the system's analytical outputs. The current dataset is sufficient to design, justify, and demonstrate the analytical pipeline but carries two limitations that are explicitly acknowledged. The 42 ad records with recorded purchases constrain the regression model to a single predictor and produced an R-squared value of 0.2658 during dataset profiling, meaning the current model explains 26.6 percent of the variation in purchases. The three non-consecutive months of campaign data prevent the application of time-series forecasting methods that require consecutive equally spaced observations. Both limitations will be directly addressed by the planned full dataset, which will provide the sample depth needed to evaluate Multiple Linear Regression and the temporal continuity needed to evaluate appropriate forecasting methods.

3.2 Conceptual Framework

The conceptual framework of the proposed system follows the Input-Process-Output (IPO) model, which illustrates how Facebook engagement data is transformed into actionable marketing and sales insights through a structured

combination of web system development and data analytics. Figure 1 presents the conceptual framework of the study.

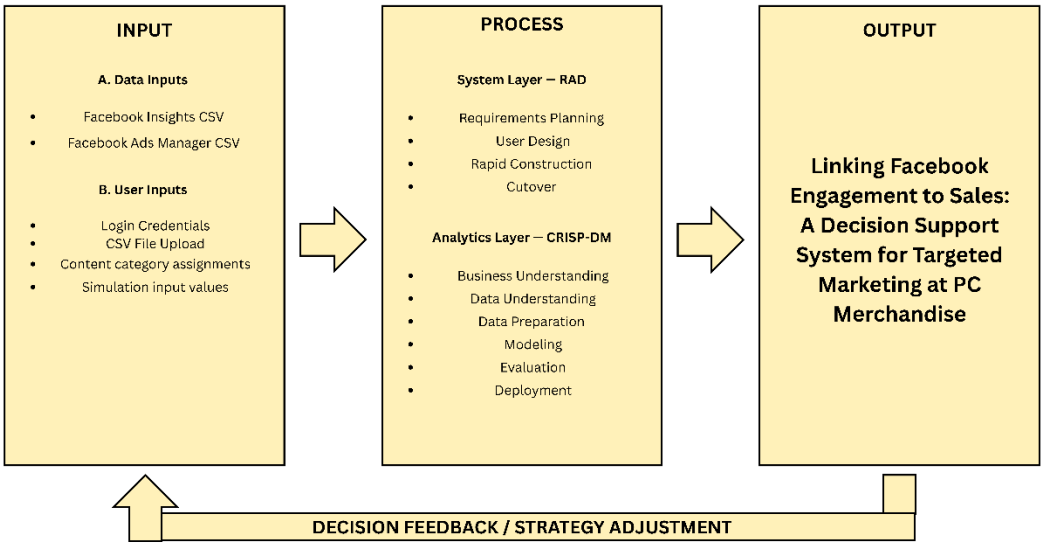

Figure 1. Conceptual Framework

The input stage consists of two categories. The first is data inputs, which comprise the two CSV file types that serve as the system's sole data sources — Facebook Insights exports containing organic page and post-level engagement metrics and Facebook Ads Manager exports containing ad-level performance data including recorded purchases. The second is user inputs, which include the login credentials users provide to access the system, the CSV files the Marketing Manager uploads through the upload interface, the content category assignments the Marketing Manager applies to organic posts and ads through the categorization

module, and the hypothetical ad spend values authorized users enter into the simulation interface.

The process stage is divided into two layers that correspond to the two methodologies guiding the development of the proposed system. The System Layer follows the Rapid Application Development methodology and covers the iterative development of the web-based components of the system across four phases — Requirements Planning, User Design, Rapid Construction, and Cutover. The Analytics Layer follows the Cross-Industry Standard Process for Data Mining and covers the data-driven components of the system across six phases — Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. The System Layer feeds into the Analytics Layer, as the web system provides the infrastructure through which data is collected, stored, and presented, while the analytics pipeline operates on that data to produce the system's core outputs. The specific activities carried out under each phase of both methodologies are discussed in detail in Section 3.3.

The output stage is represented by the proposed system in its entirety — Linking Facebook Engagement to Sales: A Decision Support System for Targeted Marketing at PC Merchandise. A feedback loop connects the output stage back to the input stage, representing the continuous and cyclical nature of the system's use in practice. Insights generated by the system — such as which ads drove the most purchases, which content categories performed best, and what the regression model predicts under hypothetical scenarios — are expected to inform

future marketing decisions at PC Merchandise, which in turn generate new Facebook engagement data and campaign records that will be uploaded into the system as new inputs. This feedback mechanism ensures that the system remains relevant and adaptive as PC Merchandise's marketing activities evolve over time.

3.3 Development Methodology

The development of the proposed system is guided by a hybrid methodology combining Rapid Application Development (RAD) and the Cross-Industry Standard Process for Data Mining (CRISP-DM). This combination is selected to address the two distinct components of the study — the web-based decision support system and the embedded analytics pipeline — each of which has different development requirements that are best served by a dedicated framework.

RAD is selected to guide the development of the web-based system. RAD is characterized by rapid prototyping, iterative development cycles, and continuous stakeholder involvement, making it particularly appropriate for a web system that must be refined progressively based on client feedback. PC Merchandise's active participation in validating system outputs — including dashboards, categorization results, and simulation interfaces — aligns directly with RAD's emphasis on user-driven development. RAD is also well-suited to the study's timeline, as its four phases — Requirements Planning, User Design, Rapid Construction, and Cutover

— allow the researchers to develop and test system modules incrementally rather than completing the entire system before any evaluation occurs.

CRISP-DM is selected to guide the analytics component of the system. CRISP-DM is the industry-standard framework for data mining and analytical modeling projects, providing a structured and reproducible process from initial business understanding through model deployment. The analytics pipeline of the proposed system — which involves dataset profiling, normality testing, correlation analysis, regression modeling, and trend analysis — maps directly onto CRISP-DM's six phases. The use of CRISP-DM ensures that the analytical methods adopted in this study are developed systematically and that each decision, from the selection of Spearman Rank Correlation to the justification of Simple Linear Regression, is grounded in a structured analytical process rather than arbitrary choice.

The integration of RAD and CRISP-DM ensures that both the software system and the analytical models will be developed in a structured, iterative, and data-driven manner. Table 5 presents the phases of the hybrid methodology, the corresponding activities under each framework, and the key deliverable produced at each phase.

Table 5. Development Methodology Phases

Phase	RAD Activities	CRISP-DM Activities	Key Deliverable
Phase 1 — Requirements Planning and Business Understanding	The researchers will gather system requirements through interviews, observation, document analysis, and dataset profiling with PC Merchandise stakeholders. User roles, system scope, and functional and non-functional requirements will be defined.	The researchers will define the business objective of linking Facebook engagement to purchase outcomes at PC Merchandise, identify the problem context, and establish the criteria for a successful analytical outcome including what the correlation, regression, and trend analysis results must demonstrate to be considered meaningful.	Requirements specification document, conceptual framework, defined analytical success criteria
Phase 2 — User Design and Data Understanding	The researchers will design the system architecture, database schema, user interface mockups, and module workflows including the upload pipeline, content categorization interface, analytics dashboard layouts,	The researchers will explore the Facebook Insights and Ads Manager datasets provided by PC Merchandise, conduct Shapiro-Wilk normality tests on all key engagement variables, compute skewness values, perform Variance	System design documents, database schema, UI mockups, normality test results, VIF analysis results, data profiling report, confirmed analytical

	and report generation structure.	Inflation Factor analysis on all candidate predictors, and assess data quality, column usability, and analytical suitability. The findings from this phase directly determine the analytical methods to be implemented in the system.	method selection
Phase 3 — Rapid Construction, Iteration 1: Core System Modules	The researchers will develop the foundational system modules including user authentication, user account management, password reset, role-based access control, CSV upload interface, data validation, and data preprocessing pipeline. These modules form the base layer that all subsequent modules depend on.	Data Preparation: The researchers will implement the data preparation logic established during Phase 2 — excluding entirely empty columns, handling missing values according to each field's analytical purpose, identifying duplicates using ad name and reporting period, standardizing date formats, and computing engagement rate for each organic post record as a derived field prior to storage.	Functional authentication, account management, upload, validation, and preprocessing modules; cleaned and structured dataset stored in MySQL database

Phase 4 — Rapid Construction, Iteration 2: Content Categorization	The researchers will develop the content categorization module including the categorization interface with permalink display for organic posts, the keyword-assisted suggestion feature for both organic posts and ads, and the keyword management interface allowing the Marketing Manager to add, edit, or remove keywords within existing categories. The predefined content categories and initial keyword sets will be finalized and loaded into the database prior to this iteration.	Data Preparation: The researchers will continue data preparation by applying keyword matching on the Title field of organic posts and the Ad name field of paid ads to generate automated category suggestions, enabling category-level aggregation of engagement and purchase metrics in subsequent analytical computations. Uncategorized records will be flagged for manual review through the categorization interface.	Functional content categorization module, categorized organic post and ads dataset, keyword sets stored in database
Phase 5 — Rapid Construction, Iteration 3: Analytics Pipeline	The researchers will integrate the analytics pipeline into the system by connecting the Django views to the Python-based analytics modules, implementing the automatic regression retraining trigger	Modeling: The researchers will implement the full analytical pipeline: Spearman Rank Correlation between engagement metrics and outcome variables for both organic post and ads datasets, Simple	Functional analytics pipeline, Spearman correlation module, regression modeling module with automatic retraining,

	following successful ads CSV uploads, and developing the What-If simulation interface.	Linear Regression using amount spent as the predictor of purchases based on VIF screening and correlation results producing the equation Purchases = 1.8168 + 0.000705 × Amount Spent, and month-over-month trend aggregation across the available campaign periods comparing total reach, impressions, spend, messaging contacts, purchases, cost per messaging contact, and cost per purchase.	trend analysis module, What-If simulation module
Phase 6 — Rapid Construction, Iteration 4: Dashboards, Reports and Simulation	The researchers will develop the role-appropriate dashboards for the Marketing Manager, Sales Director, and Business Owner, the campaign performance ranking interface with sortable metrics and high-performing and underperforming labels, the report generation module,	Evaluation: The researchers will evaluate all analytical outputs produced by the system against the success criteria defined in Phase 1. This includes assessing the R-squared value of 0.2658 from the regression model, verifying that Spearman	Functional role-appropriate dashboards, campaign performance ranking interface, report generation module, complete What-If simulation interface,

	and the complete What-If simulation interface displaying regression equation and baseline comparison.	correlation coefficients and p-values are correctly computed and displayed, confirming that trend aggregations accurately reflect the stored data, and ensuring that simulation outputs correctly apply the trained regression equation to user inputs.	model evaluation results
Phase 7 — Cutover and Deployment	The researchers will conduct system testing covering functional correctness, role-based access control enforcement, upload validation accuracy, preprocessing consistency, and analytical output integrity. User acceptance testing will be conducted with PC Merchandise stakeholders. Upon successful testing, the system will be deployed on the local host machine at PC Merchandise using Cloudflare Tunnel,	The researchers will finalize the deployment of the analytical models within the system, confirming that the Spearman correlation, regression modeling, trend analysis, and simulation modules all produce consistent and accurate results under the deployed environment. The researchers will also document the current model's limitations — specifically the R-squared value of	Fully tested and deployed system, user acceptance testing report, user guidance documentation, deployed analytical pipeline, model limitation documentation and expansion plan

	and user guidance will be provided to all three user roles.	0.2658 and the three non-consecutive months of trend data — and outline the plan for revisiting the Modeling phase upon acquisition of the complete eight-month dataset from PC Merchandise. Lastly, the flag-and-suggest category detection mechanism is documented as a planned enhancement to be evaluated upon acquisition of the complete eight-month dataset.	

The table reflects the parallel structure of RAD and CRISP-DM at every phase. The four iterations presented are the planned development sequence based on the logical dependencies between system modules. In practice, RAD allows iterations to be added or repeated based on testing outcomes and stakeholder feedback, and the iteration structure may be adjusted accordingly during actual development.

It is important to note that the analytical findings produced during the Data Understanding phase — specifically the Shapiro-Wilk normality test results, VIF

analysis, and Spearman correlation results — are based on the initial dataset provided by PC Merchandise for the purposes of this proposal. Upon acquisition of the complete eight-month dataset following proposal approval, the researchers will revisit the Data Understanding and Modeling phases to evaluate whether the expanded data supports the application of Multiple Linear Regression with additional VIF-screened predictors and appropriate time-series forecasting methods. The modular design of the analytics pipeline ensures that such expansions can be implemented without requiring structural changes to the system architecture.

3.4 System Design

This section presents the structural and architectural design of the proposed Linking Facebook Engagement to Sales: A Decision Support System for Targeted Marketing at PC Merchandise. It provides a visual blueprint that translates the system requirements defined in Section 3.1 into a coherent and organized system structure. The diagrams presented in this section illustrate the system's boundaries, data flows, user interactions, database structure, component organization, process logic, and user interface design. Together, these diagrams serve as the foundation for the system's development, ensuring that all modules are aligned with the identified functional and non-functional requirements.

3.4.1 Context Diagram

The Context Diagram, also known as the Level 0 Data Flow Diagram, presents the proposed system as a single central process following the Gane and Sarson notation. It establishes the system's boundaries by showing how the proposed system interacts with its three entities — the Marketing Manager, the Sales Director, and the Business Owner — without detailing the internal processes that occur within the system. Figure 2 presents the Context Diagram of the proposed system..

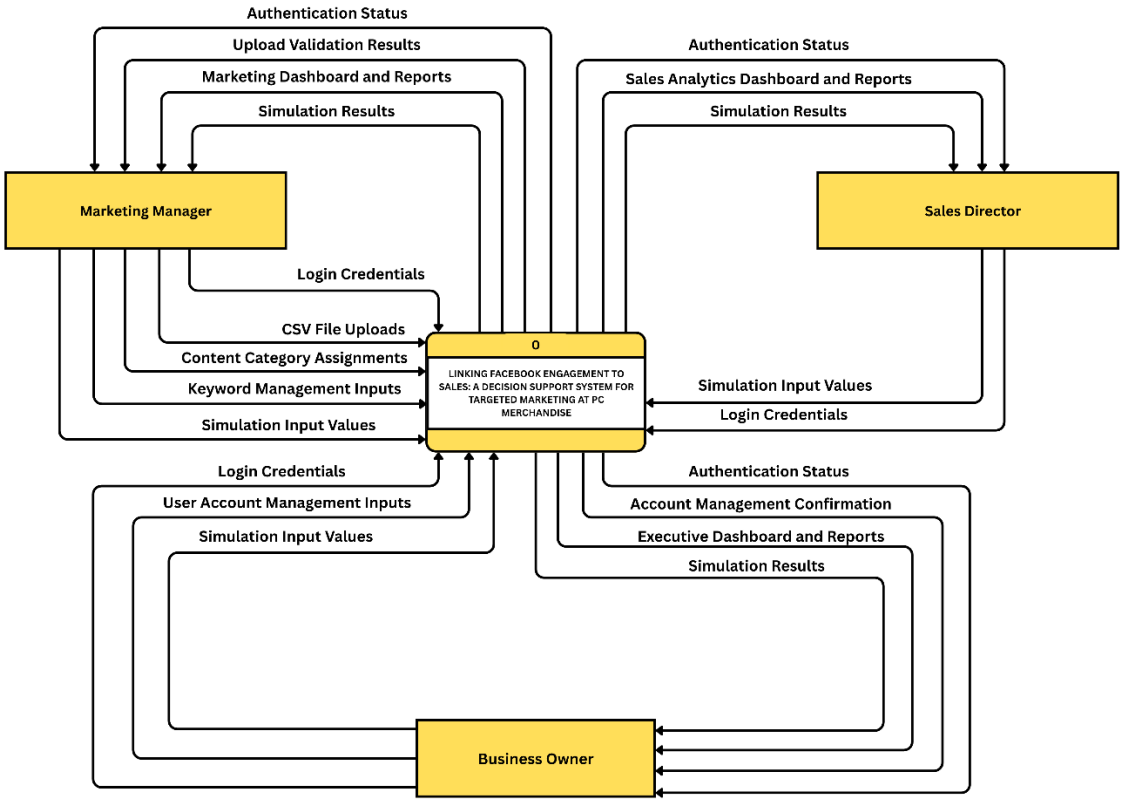

Figure 2. Context Diagram – Level 0 DFD

As shown in Figure 2, three entities interact with the central process, each corresponding to a distinct user role defined in the system's role-based access control design.

The Marketing Manager is the sole data operator of the system. Login credentials, CSV file uploads containing Facebook Insights and Ads Manager data, content category assignments, and simulation input values are sent into the system. In return, the system provides authentication status, upload validation results, marketing dashboard and reports, and simulation results. The Sales Director is an analytical consumer whose interaction with the system is accessing outputs relevant to sales performance monitoring and running what-if analysis. Login credentials are sent into the system and authentication status and sales analytics dashboard and reports are received in return. The Business Owner is the strategic and administrative user of the system. Login credentials, user account management inputs, and simulation input values are sent into the system. In return, the system provides authentication status, executive dashboard and reports, and simulation results. The internal processes through which the system handles these data flows and produces its outputs are discussed in detail in Section 3.4.2.

3.4.2 Data Flow Diagram – Level 1

The Level 1 Data Flow Diagram expands the single central process from the Context Diagram into seven distinct processes, providing a more detailed view of how data moves through the proposed system. All data stores shown in the diagram reside within a single MySQL database and represent logical groupings of closely related tables that are consistently accessed together by the same processes. Figure 3 presents the Level 1 Data Flow Diagram of the proposed

system.

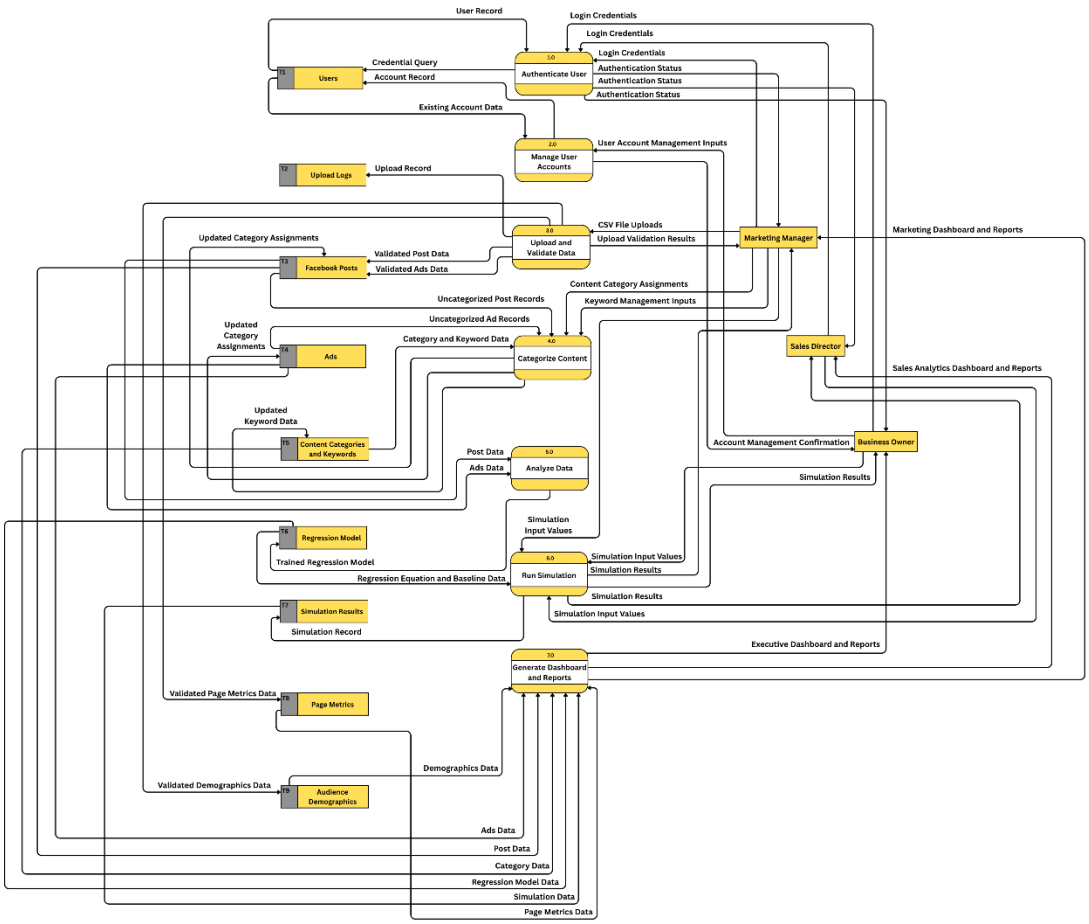

Figure 3. Data Flow Diagram – Level 1

As shown in Figure 3, the system is composed of seven processes and nine data stores, with three entities interacting with the system. Process 1.0 handles user authentication and establishes the user role for all subsequent processes. Process 2.0 manages user account creation, editing, and deactivation exclusively for the Business Owner. Process 3.0 handles the intake, validation, and

preprocessing of all CSV uploads from the Marketing Manager, distributing validated organic post data to T3, validated ads data to T4, validated daily page-level metrics to T8, and validated follower demographic snapshot data to T9, while recording every upload in T2 regardless of outcome and passing a trigger to Process 5.0 when new purchase records are detected. Process 4.0 manages content categorization for both organic posts and ads based on category assignments and keyword management inputs from the Marketing Manager, reading from and writing back to T3, T4, and T5. Process 5.0 performs all statistical computations on data read from T3 and T4 using GROUP BY queries, writing the trained regression model parameters to T6 each time retraining is triggered by Process 3.0. Process 6.0 applies the stored regression equation to hypothetical inputs from any of the three entities, records each simulation run in T7, and returns projected results alongside previously run simulation history. Process 7.0 reads from all relevant data stores and generates role-appropriate dashboard and report outputs for each of the three entities based on the authenticated session established in Process 1.0.

3.4.3 Use Case Diagram

The Use Case Diagram presents the functional interactions between the system's three actors and its core features, defining what the system does for each user role. Figure 4 presents the Use Case Diagram of the proposed system.

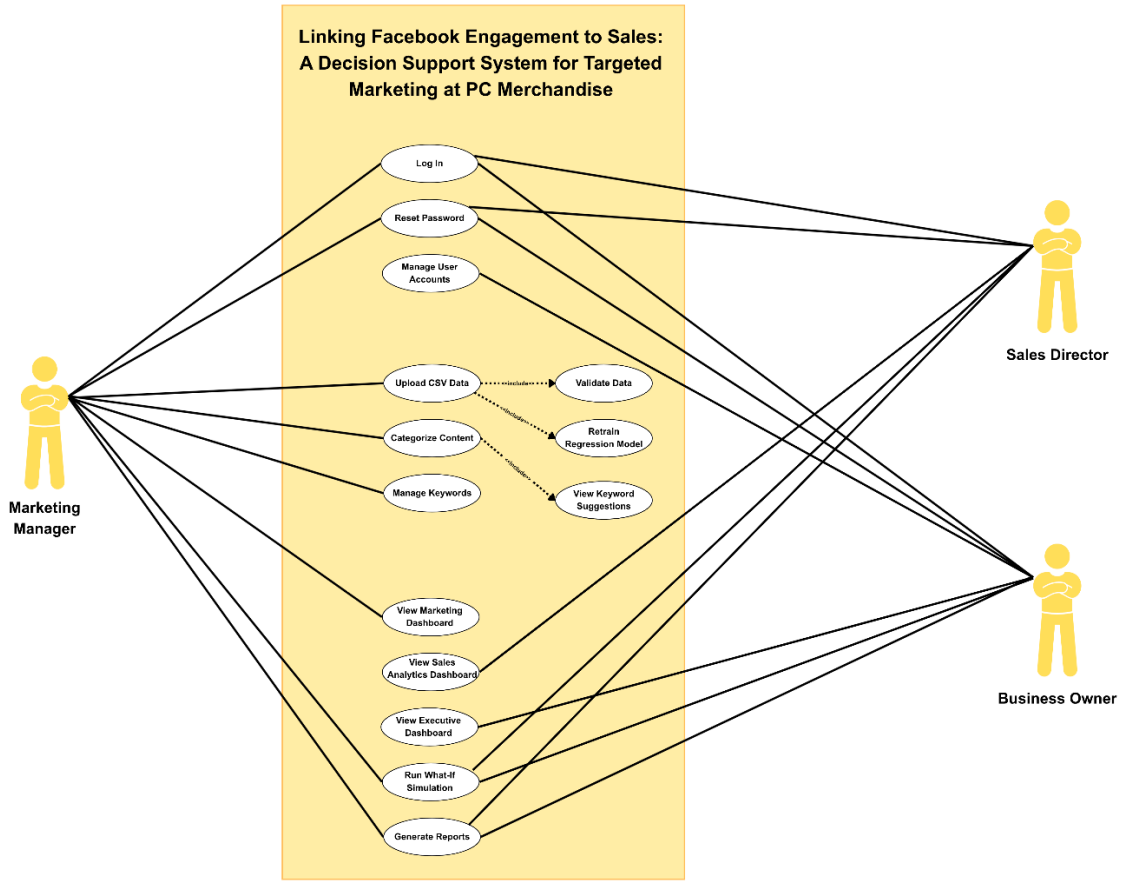

Figure 4. Use Case Diagram

As shown in Figure 4, three actors interact with the system — the Marketing Manager, the Sales Director, and the Business Owner — each connected to the use cases relevant to their assigned role.

All three actors share access to Log In, Reset Password, Run What-If Simulation, and Generate Reports. The Marketing Manager, as the sole data operator, additionally accesses Upload CSV Data, Categorize Content, and Manage Keywords, and views the Marketing Dashboard. The Sales Director accesses the Sales Analytics Dashboard. The Business Owner accesses the Executive Dashboard and is the only actor who can perform Manage User Accounts. The three role-specific dashboard use cases reflect the system's role-based access control design, where each role receives a distinct set of analytical outputs appropriate to its responsibilities.

Three include relationships are present in the diagram. Upload CSV Data includes Validate Data, meaning validation runs automatically with every upload. Upload CSV Data also includes Retrain Regression Model, meaning the regression model retrains automatically when a new ads upload adds purchase records. Categorize Content includes View Keyword Suggestions, meaning automated keyword-based category suggestions are generated each time the categorization interface is opened. These included use cases are not initiated directly by any actor but are triggered internally as part of their respective base use cases.

3.4.4 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram presents the logical structure of the database underlying the proposed system. It defines the tables that store the system's data, the attributes associated with each table, and the relationships that connect them. The diagram follows Crow's Foot notation and serves as the blueprint for the system's single MySQL database. Figure 5 presents the Entity Relationship Diagram of the proposed system.

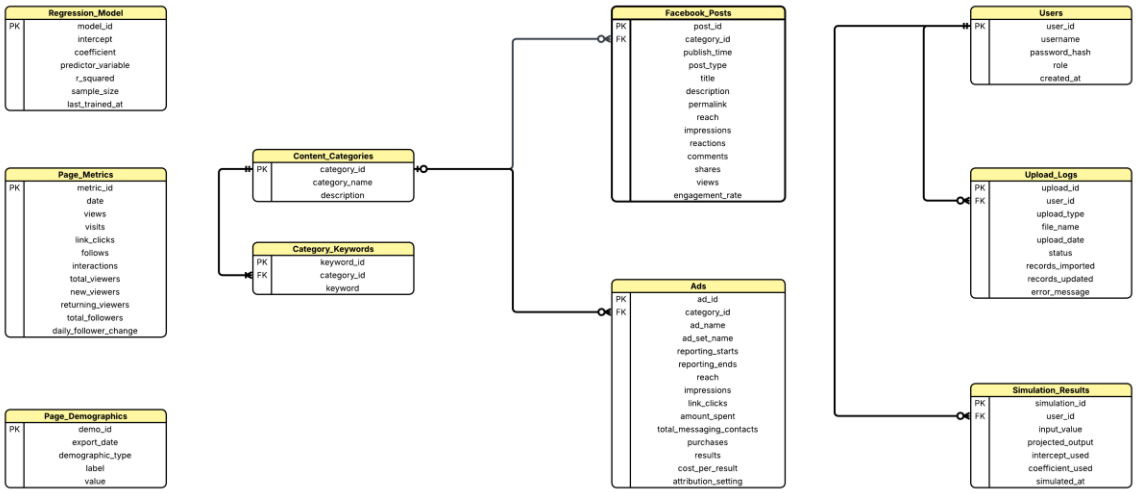

Figure 5. Entity Relationship Diagram (ERD)

As shown in Figure 5, the database is composed of ten tables organized around the core functional areas of the system. T1 Users and T2 Upload_Logs support user authentication, account management, and upload auditing. T8 Simulation_Results stores the history of all simulation runs and is linked to T1 Users so each simulation is attributed to the user who ran it. T3 Facebook_Posts and T4 Ads store the core analytical datasets imported from Facebook Insights and Facebook Ads Manager CSV exports respectively. T5 Content_Categories

and T6 Category_Keywords support the content categorization module — T5 stores the predefined category names and T6 stores the associated keyword sets, with each keyword linked to exactly one parent category. Both T3 and T4 contain a nullable foreign key referencing T5, reflecting the fact that posts and ads may exist in the system without a category assigned until the Marketing Manager completes categorization. T7 Regression_Model stores the parameters of the most recently trained regression model and stands alone with no relationships since it is written and read exclusively by the application layer. T9 Page_Metrics and T10 Page_Demographics store page-level data imported from the page-level CSV exports and likewise stand alone as display-only tables that do not participate in the core analytical computations.

3.4.5 System Architecture Diagram

The System Architecture Diagram presents the overall structure of the proposed system, illustrating how its components are organized across three distinct layers and how data flows between them. The diagram follows a three-tier architecture composed of the Presentation Layer, the Application Layer, and the Data Layer. All three layers run on a single designated host machine at PC Merchandise and are made internet-accessible to authorized users through Cloudflare Tunnel, which establishes a secure encrypted HTTPS connection

between the host machine and any device the authorized users choose to access the system from. Figure 6 presents the System Architecture Diagram of the proposed system.

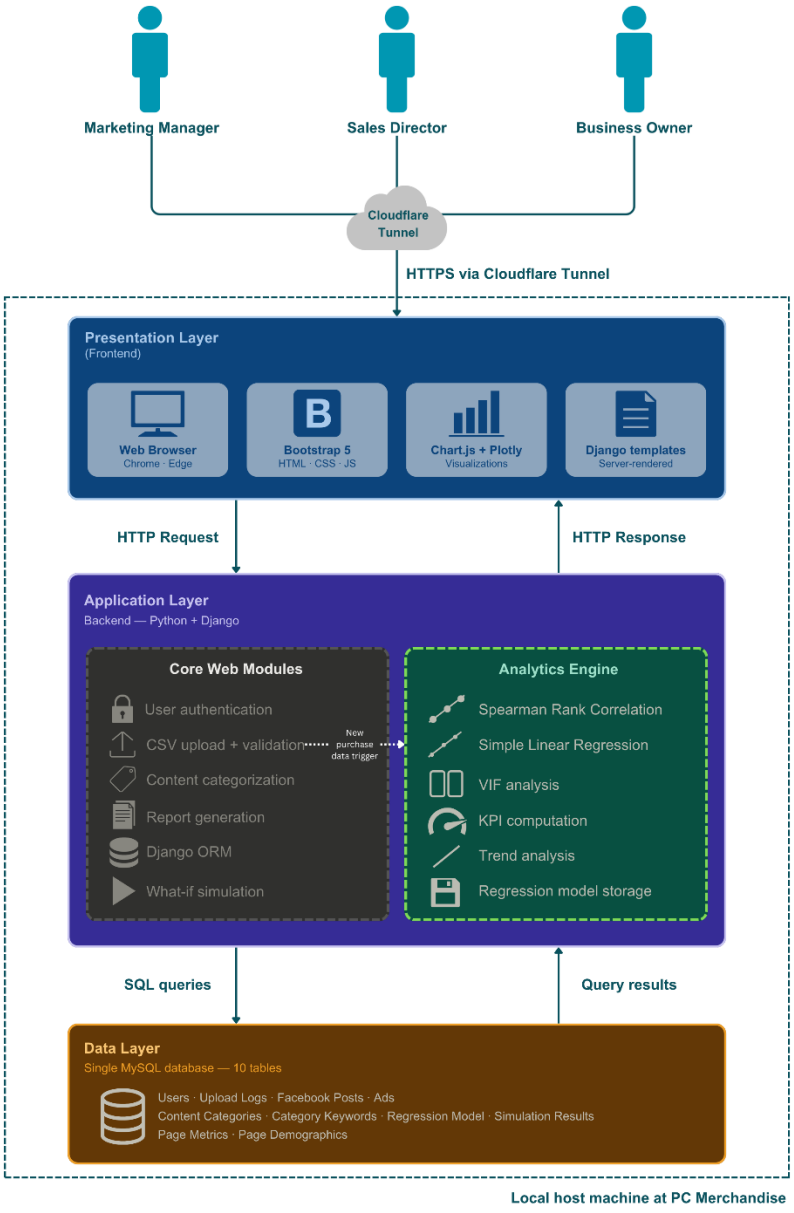

Figure 6. System Architecture Diagram

As shown in Figure 6, the three-tier architecture separates the user interface, the business logic and analytics processing, and the data storage into clearly defined layers, ensuring that the system is modular and maintainable in alignment with NFR-08.

The Presentation Layer represents the frontend of the system — the part that users directly interact with through a standard web browser on any internet-connected device. This layer is composed of Bootstrap 5, which handles the responsive layout and styling of all interface components; Chart.js, which renders standard dashboard charts including bar charts, line graphs, and trend visualizations; Plotly, which renders interactive analytical visualizations including the correlation heatmap and regression scatter plot; and Django HTML Templates, which are server-rendered pages returned by the backend in response to each user request. All three user roles access the system exclusively through this layer using a web browser without requiring any additional software installation on their own devices.

The Application Layer is the backend of the system and contains all business logic, data processing, and analytical computation. It is built entirely on Python using Django as the web framework and is divided into two groups of components. The first group — the Core Web Modules — handles the operational functions of the system including user authentication, CSV upload and validation,

content categorization, report generation, the Django ORM for database communication, and the What-If simulation interface. The second group — the Analytics Engine — houses all Python-based statistical modules including Spearman Rank Correlation using scipy, Simple Linear Regression and VIF analysis using scikit-learn, KPI computation and trend analysis using pandas, and regression model parameter storage. The Analytics Engine is triggered automatically by the upload process when new purchase records are detected, ensuring the regression model always reflects the most current dataset without requiring manual intervention from any user.

The Data Layer consists of the single MySQL relational database that stores all persistent data used by the system across ten tables — Users, Upload Logs, Facebook Posts, Ads, Content Categories, Category Keywords, Regression Model, Simulation Results, Page Metrics, and Page Demographics. All data in this layer is accessed exclusively through the Django ORM in the Application Layer, ensuring no component in the Presentation Layer communicates directly with the database.

The three-tier architecture ensures a clear separation of concerns between the user interface, the processing and analytics logic, and the data storage. This separation supports the maintainability and scalability requirements defined in NFR-07 and NFR-08, as individual components within each layer can be updated or extended without requiring changes to the other layers.

3.4.6 Flowchart

This section presents the process flowcharts for the major workflows of the proposed system. Figure 7.1 illustrates the System Access workflow, covering user authentication, role-based redirection, and action selection. Figure 7.2 illustrates the Data Preprocessing workflow, covering CSV upload, validation, cleaning, and storage. Figure 8 illustrates the Algorithm Execution workflow, covering content categorization, statistical analysis, regression modeling, trend analysis, simulation, and dashboard and report generation. Together these three flowcharts document the step-by-step logic and decision points that drive the system's core access, data management, and analytical capabilities.

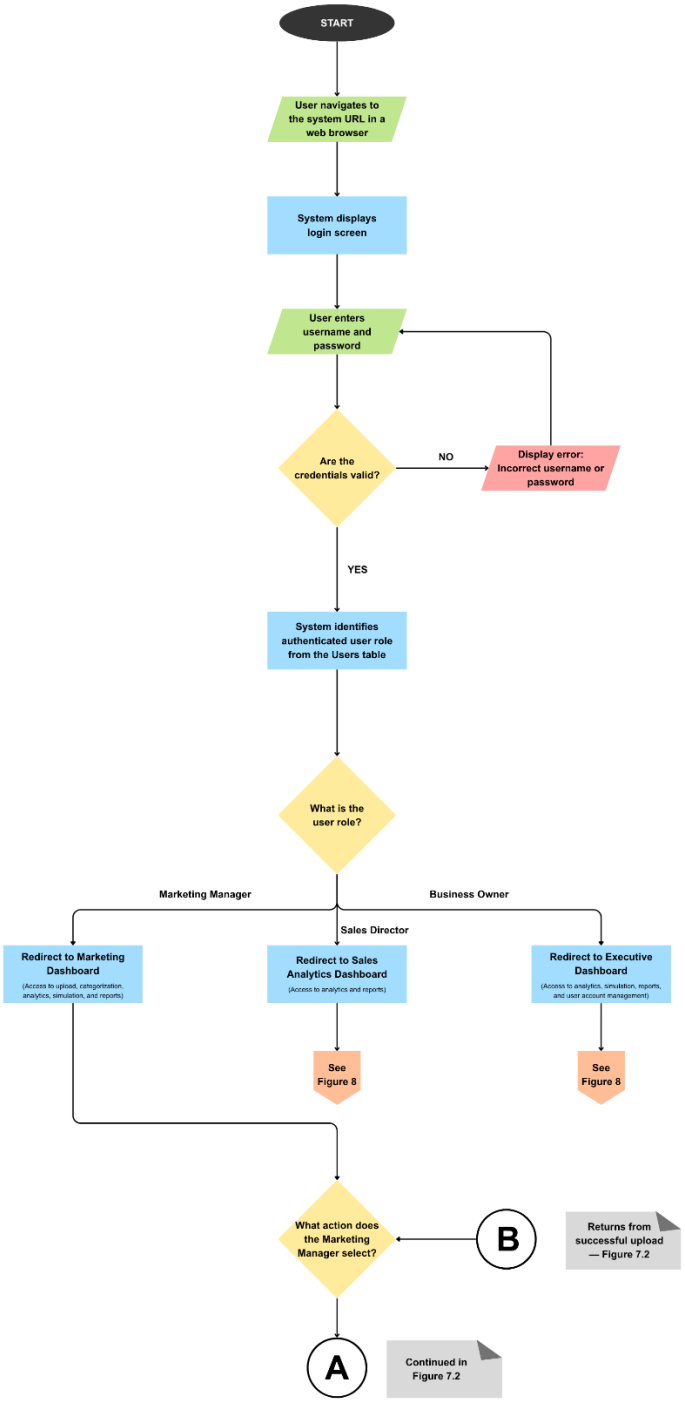

Figure 7.1. System Access Flowchart

As shown in Figure 7.1, the flow begins when a user navigates to the system URL and is presented with the login screen. Credentials are validated server-side against the Users table, and an error message is displayed with a return to the login screen if authentication fails. Upon successful authentication, the system identifies the user role and redirects the Marketing Manager to the Marketing Dashboard, the Sales Director to the Sales Analytics Dashboard, and the Business Owner to the Executive Dashboard. The Sales Director and Business Owner proceed directly to the analytical workflow described in Figure 8. The Marketing Manager is presented with an action selection where the flow either continues into the CSV upload and preprocessing workflow in Figure 7.2, proceeds to the analytical workflow in Figure 8, or terminates the session through logout.

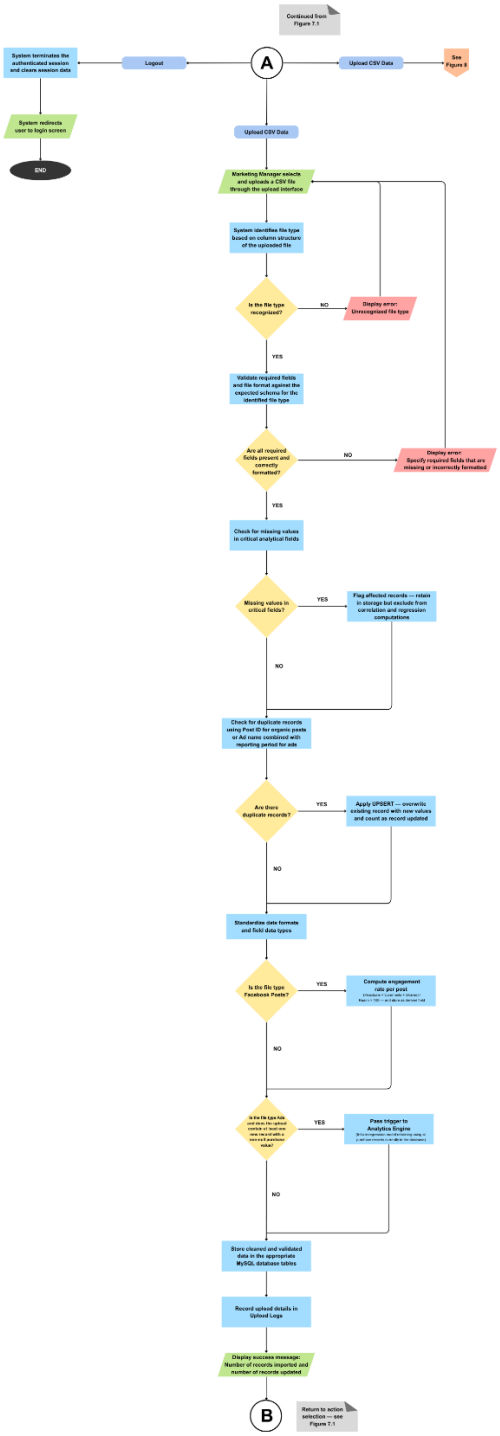

Figure 7.2. Data Preprocessing Flowchart

As shown in Figure 7.2, the flow begins when the Marketing Manager selects and uploads a CSV file. The system identifies the file type based on column structure — an unrecognized file type returns an error and redirects the Marketing Manager to re-upload. Required fields and file format are then validated against the expected schema for the identified file type — invalid files return a specific error and redirect to re-upload. Records with missing values in critical analytical fields are flagged and retained in storage but excluded from analytical computations. Duplicate records are resolved through UPSERT — overwriting existing records with updated values and counting them as updated rather than imported. Date formats and field data types are then standardized across the dataset. If the file type is Facebook Posts, engagement rate is computed for each record as a derived field prior to storage. If the file type is Ads and the upload contains at least one new record with a non-null purchase value, a trigger is passed to the Analytics Engine to initiate regression model retraining. The cleaned and validated data is then stored in the appropriate MySQL database tables, the upload details are recorded in Upload Logs, and a success message is displayed to the Marketing Manager showing the number of records imported and updated. The flow then returns to the action selection in Figure 7.1 where the Marketing Manager may perform another action.